

TATEYAMA, Japan

WMO: 476720

Lat: 34.9869N

Lon: 139.8653E

Elev: 23

StdP: 14.68

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	29.0	31.0	11.4	9.8	41.5	14.7	11.6	41.9	25.2	50.9	22.9	50.5	4.1	160	0.408

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	11.3	89.0	78.6	87.5	78.2	86.0	77.7	80.2	85.8	79.5	84.8	78.7	83.9	9.4	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
78.8	149.4	83.3	78.0	145.4	82.8	77.2	141.5	82.4	43.9	86.0	43.1	85.0	42.3	84.1	83.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min		Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.2	18.7	16.3	DB	25.0	92.5	1.7	1.7	23.7	93.7	22.7	94.7	21.8	95.6	20.5	96.8
			WB	23.1	81.5	1.7	1.0	21.8	82.3	20.8	82.9	19.8	83.5	18.6	84.2

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	61.3	43.1	44.1	49.5	57.6	65.4	71.0	78.2	80.9	75.1	65.9	56.4	47.8
	DBStd	13.70	4.38	5.10	5.77	5.65	4.09	3.80	4.01	3.40	5.03	5.00	5.31	5.19
	HDD50	604	222	180	81	8	0	0	0	0	0	0	7	106
	HDD65	2865	680	586	479	227	45	3	0	0	1	48	262	534
	CDD50	4746	7	15	67	237	476	629	873	958	754	494	198	38
	CDD65	1532	0	0	0	6	57	181	408	493	306	77	4	0
	CDH74	11406	0	0	0	2	94	585	3284	4933	2226	280	2	0
	CDH80	3010	0	0	0	0	3	47	832	1600	510	18	0	0
	WSAvg	6.8	6.7	6.5	7.5	7.8	7.3	6.8	7.5	7.1	6.4	6.0	5.8	6.5
Precipitation	PrecAvg	75.5	3.5	3.9	6.4	6.5	6.7	8.7	6.9	4.8	9.0	9.3	6.1	3.5
	PrecMax	93.8	9.0	7.4	11.5	12.6	12.0	16.7	18.0	13.6	18.2	25.4	18.6	10.2
	PrecMin	49.9	0.0	0.7	2.2	3.3	3.3	3.5	0.2	0.1	2.7	2.0	1.7	0.2
	PrecStd	11.9	2.2	1.9	2.3	2.4	2.2	3.5	4.8	3.6	4.0	5.6	3.7	2.2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	62.5	65.2	68.3	73.3	79.2	84.3	90.5	91.8	88.8	82.3	73.6	68.6
		MCWB	56.8	58.9	58.8	63.9	67.0	74.4	78.9	79.6	77.4	72.3	65.0	63.3
	2%	DB	59.0	61.6	65.5	71.0	76.3	81.0	88.1	89.7	86.7	79.3	71.1	64.6
		MCWB	52.0	55.6	57.8	62.8	66.4	72.5	78.3	78.9	77.3	71.3	63.9	57.5
	5%	DB	56.0	58.2	63.3	69.3	74.6	79.0	86.5	88.3	85.0	77.1	69.2	61.6
		MCWB	47.9	51.5	56.5	61.3	65.6	71.3	77.8	78.7	76.7	70.2	62.0	53.6
	10%	DB	53.3	55.0	61.2	67.4	73.0	77.3	84.8	87.0	83.1	75.0	67.1	58.5
		MCWB	44.8	47.3	54.2	60.0	65.2	70.7	77.1	78.3	75.7	67.8	60.0	50.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	58.5	60.8	62.6	68.8	72.1	76.8	80.6	81.7	79.9	76.0	68.7	65.0
		MCDB	61.2	63.6	64.7	70.7	73.9	81.3	87.1	88.3	84.9	78.9	71.2	67.2
	2%	WB	53.1	56.8	60.1	65.4	70.0	75.1	79.4	80.7	78.9	73.6	65.9	58.9
		MCDB	57.9	60.5	63.9	68.7	73.2	78.6	85.3	86.5	83.7	77.5	69.6	63.1
	5%	WB	49.0	52.5	58.0	63.4	68.4	73.7	78.7	80.0	78.0	71.4	63.6	54.8
		MCDB	54.7	57.4	62.5	67.5	72.2	77.2	84.2	85.5	82.8	75.9	67.7	60.1
	10%	WB	45.8	48.5	55.1	61.4	67.0	72.4	78.0	79.3	77.1	69.1	61.2	51.6
		MCDB	51.9	53.8	60.1	66.0	71.2	75.8	83.0	84.5	81.9	73.8	65.6	57.0
Mean Daily Temperature Range	5% DB	MDBR	17.0	16.4	15.8	15.1	13.4	10.9	10.4	11.3	12.2	13.8	15.7	16.8
		MCDBR	19.4	19.1	16.8	16.1	15.3	13.0	12.4	12.5	12.9	14.9	17.6	19.1
		MCWBR	14.6	15.3	12.6	10.0	8.0	6.2	5.0	4.8	6.0	9.2	11.9	14.0
	5% WB	MCDBR	18.1	17.5	15.3	12.2	11.8	9.6	11.2	11.1	11.4	12.7	14.7	17.5
		MCWBR	15.0	15.7	13.8	9.7	8.1	5.7	4.9	4.7	6.1	9.5	11.9	14.7
Clear-Sky Solar Irradiance	taub		0.388	0.438	0.500	0.543	0.557	0.572	0.568	0.550	0.499	0.482	0.432	0.389
	taud		2.233	2.090	1.941	1.882	1.910	1.948	2.004	2.046	2.160	2.121	2.178	2.237
	Ebn at Noon		251	250	244	241	239	235	235	236	243	233	233	241
	Edh at Noon		35	45	56	62	62	59	56	53	45	43	36	33
All-Sky Solar Radiation	RadAvg		832	986	1259	1491	1625	1428	1641	1665	1277	957	775	739
	RadStd		67	110	89	156	170	142	266	140	110	100	70	62

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	+0.74	N/A	N/A	N/A	N/A
(47)	Regional (36 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page